Received: 24 April 2026 | Accepted: 21 September 2026

DOI: 10.1002/ajb2.70274

ON THE NATURE OF THINGS

Botany

Beyond the tangled bank: An evolving meta-ecosystems framework to integrate ecological and evolutionary perspectives

Brandon M. Lind[1,2] | **Ian N. C. Shuman**[3,4] | **Linda A. Deegan**[5] | **Mary Heskel**[6] | **Mark C. Urban**[1,7] | **Jill L. Wegrzyn**[1,2] | **Peter A. H. Westley**[8]

[1]Department of Ecology & Evolutionary Biology, University of Connecticut, Storrs, Connecticut 06269, USA

[2]Institute for Systems Genomics, University of Connecticut, Storrs, Connecticut 06269, USA

[3]Department of Earth and Environmental Sciences, Columbia University, New York, New York 10027, USA

[4]Lamont-Doherty Earth Observatory, Columbia University, Palisades, New York 10964, USA

[5]Woodwell Climate Research Center, Falmouth, Massachusetts 02540, USA

[6]Department of Biology, Macalester College, Saint Paul, Minnesota 55105, USA

[7]Center of Biological Risk, University of Connecticut, Storrs, Connecticut 06269, USA

[8]College of Fisheries and Ocean Sciences, University of Alaska Fairbanks, Fairbanks, Alaska 99775, USA

Correspondence

Brandon M. Lind, Department of Ecology & Evolutionary Biology, University of Connecticut, Storrs, Connecticut 06269, USA.
Email: lind.brandon.m@gmail.com

KEYWORDS

adaptive evolution, cross-ecosystem subsidies, eco-evolutionary dynamics, ecosystem coupling, global change biology, meta-ecosystems

Funding information

Directorate for Biological Sciences, Grant/Award Number: 2320675

> "It is interesting to contemplate a tangled bank, clothed with many plants of many kinds, with birds singing on the bushes, with various insects flitting about, and with worms crawling through the damp earth, and to reflect that these elaborately constructed forms, so different from each other, and dependent upon each other in so complex a manner, have all been produced by laws acting around us."
>
> Charles Darwin, *On the Origin of Species*

Darwin's tangled bank (Darwin, 1869) describes life as a network of interdependent organisms shaped by eco-evolutionary processes. This metaphor has gained renewed relevance in the 21st century, as rapid environmental and land-use changes drive ecological reorganization at multiple biological and spatial scales. Yet Darwin's metaphor was largely local—a single bank. While species interactions and selection pressures can take place at relatively local spatial scales, energy, resources, and organisms move across atmospheric, terrestrial, and aquatic boundaries, connecting ecosystems into larger interacting meta-ecosystem networks (Loreau et al., 2003). The meta-ecosystem framework combines concepts from meta-population and meta-community theory—which describe spatially structured populations of one or more species connected by the flow of individuals and their interactions—with those from eco-system ecology—through the tracking of the sources, sinks, and fluxes of energy and materials. Together, it extends beyond the study of individual systems to describe the dynamics of *multiple* ecosystems connected by the "flows of energy, materials, and organisms across ecosystem boundaries" (Loreau et al., 2003). This conceptual foundation has provided a powerful framework to understand emergent properties of these networks that arise from spatiotemporal

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coupling within and across constituent components. For instance, Massol et al. (2011) theoretically demonstrated that interactions between flows of materials and organisms can affect population persistence and the stability of connected systems in ways that are not apparent when community and ecosystem processes are considered in isolation. Indeed, subsequent empirical studies have found that fluxes across system boundaries can fundamentally change community structure, population dynamics, and ecosystem properties and functions (Gounand et al., 2018). Yet, despite its strengths, meta-ecosystem theory has remained predominantly focused on ecological dynamics, largely ignoring intraspecific genetic variation that provides an additional subsidy—the substrate for evolution by natural selection. Therefore, the very processes implicit in Darwin's *On the Origin of Species* and his tangled bank metaphor are rarely considered within the meta-ecosystem framework.

THE EVOLVING META-ECOSYSTEM FRAMEWORK

Over the last several decades, there has been increasing recognition that adaptive evolution can take place over a few generations, on similar timescales as ecological change, and over fine, microgeographic scales (Carroll et al., 2007; Richardson et al., 2014). Furthermore, the traits that respond to natural selection—such as phenology, growth, stress tolerance, resource allocation, and reproduction—can influence ecosystem processes, such as productivity, decomposition, carbon sequestration, and even landscape morphology and geophysical properties (Dong et al., 2024).

Despite the exclusion of evolutionary dynamics within theoretical meta-ecosystem models, there is ample evidence that evolutionary change can cause ecological change across ecosystem boundaries. For example, black cottonwood (*Populus trichocarpa* Torr. & A.Gray ex Hook.; Salicaceae) grows in riparian zones and delivers annual pulses of leaf litter into streams and rivers, providing a major subsidy that supports aquatic food webs. Common gardens have shown that genetic variation in leaf senescence alters decomposition rates and aquatic invertebrate richness (Rodriguez-Cabal et al., 2017). Further, studies in other *Populus* species have shown that genetic variation in leaf chemistry also influences decomposition, invertebrate assemblages, and soil net nitrogen (Schweitzer et al., 2008). These and other examples—such as recent body size declines in Pacific salmon and subsequent decreases in marine-derived nutrient transport to freshwater systems (Oke et al., 2020)—illustrate that variation and contemporary changes in key traits can alter meta-ecosystem dynamics through the flow of material, energy, and organisms across ecosystem boundaries.

In the face of global change and growing recognition of the potential for evolutionary influence on meta-ecosystem coupling, we must ask: (1) How does adaptive evolution shape mechanistic responses to environmental change at multiple levels of biological organization—from genes to ecosystems? And (2) How do these interacting processes determine the resilience of meta-ecosystem functions? To address these questions, meta-ecosystems cannot be treated simply as networks of habitats connected by flows, but instead as dynamic systems with connections that are mediated by organismal evolution and ecological feedbacks—an *evolving* meta-ecosystem (Figure 1).

EXPERIMENTAL DESIGNS AND THE NECESSARY DATA TO UNDERSTAND EVOLVING META-ECOSYSTEM DYNAMICS

Determining whether evolutionary responses to environmental change have consequential impacts on spatiotemporal meta-ecosystem dynamics requires quantifying and linking: (1) underlying relationships between environmental change and the evolutionary responses of heritable, genetically-based traits (including heritable plasticity); (2) the extent to which those traits directly or indirectly impact cross-boundary flows; and (3) how changes in the properties of these flows (e.g., the direction, magnitude, and spatiotemporal variation and synchrony) caused by evolutionary responses impact meta-ecosystem function and stability—for example, in carbon or nutrient balance, trophic subsidies, connectivity, feedback dynamics, or the resilience of coupled meta-ecosystem functions to disturbance or environmental change.

Temporal and contemporary approaches offer complementary perspectives for understanding evolving meta-ecosystems. For instance, retrospective or long-term observations and experiments can each be used to demonstrate evolutionary change. Moreover, evidence of evolutionary change that coincides with environmental change can provide evidence for an evolutionary response. Resurrection studies of seeds collected from natural or human-mediated seed banks, for example, can reveal how ecologically relevant traits have shifted over periods of changing temperature, hydrology, or nutrient regimes (Christie et al., 2023). Herbarium and museum collections can also provide material for time-series genomics data, which can be used to quantify signals of adaptive responses at the genetic level. Similarly, long-term ecological research and monitoring networks can capture both ecological and evolutionary change (Cocciardi et al., 2024).

Contemporary organismal approaches complement these temporal perspectives. Common gardens or long-term transplant experiments along ecological gradients can quantify the heritable and environmental components of phenotypic variation—including genetic correlations (e.g., trade-offs)—for traits that underlie fitness, mediate meta-ecosystem flows, or both (Blum et al., 2021). These experiments can also disentangle heritable plasticity from

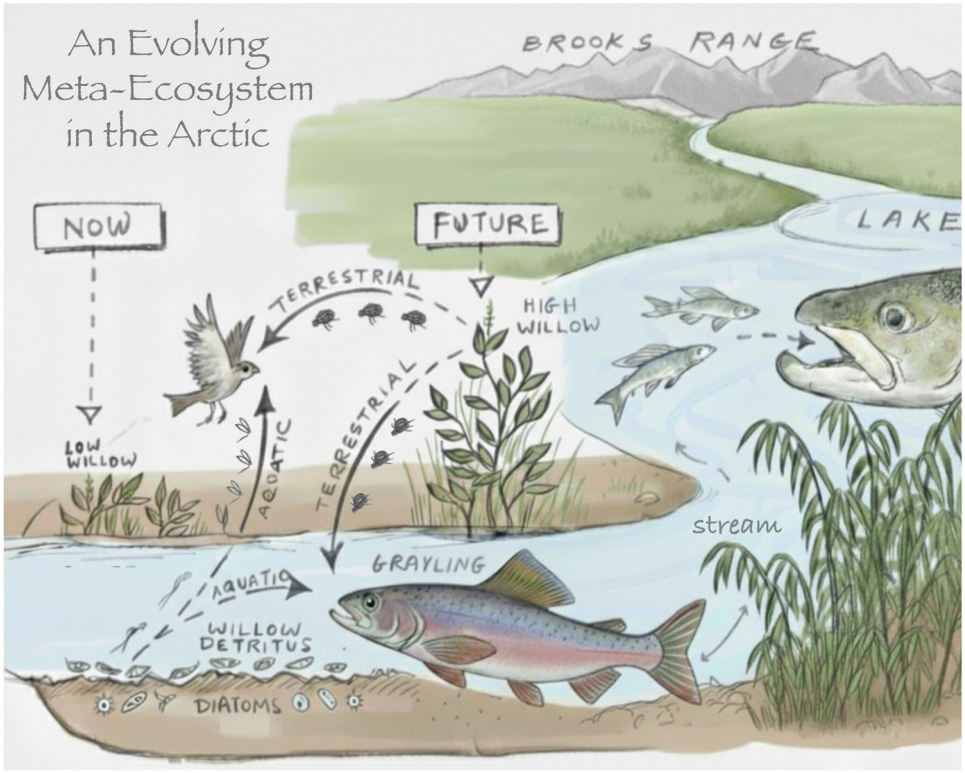


FIGURE 1 Evolutionary processes impact cross-boundary flows of energy, materials, and organisms in an evolving meta-ecosystem. This figure depicts a tangled bank within an Arctic stream-riparian meta-ecosystem network, where contemporary climate change has driven expansion and increased growth of a dominant willow species (*Salix alaxensis*). These evolutionary responses by *S. alaxensis* and other Arctic shrubs are expected to continue and become more prevalent in the future, with the potential to dramatically alter ecological and evolutionary dynamics within and across terrestrial and aquatic ecosystems. In this evolving meta-ecosystem, cross-boundary flows—including willow leaf litter, emerging insects, and movement of Arctic grayling (*Thymallus arcticus*) and white-crowned sparrow (*Zonotrichia leucophrys*)—influence, interact with, and feed back into evolutionary responses of plant phenology and growth, the population dynamics of insect, bird, and fish species, species interactions within and across ecosystem boundaries, and nutrient cycling regimes. Likewise, each of these aspects can influence, interact with, and feed back into other properties within and across boundaries of the meta-ecosystem. Here, transplant experiments and genomic studies can establish signals of local adaptation and contemporary adaptive evolution (e.g., selective sweeps) in these species. The impact of evolutionary responses from increased shrub growth on trophic subsidies across ecosystem boundaries could be quantified through leaf litter measurements (including chemical analysis), the resulting dynamics of aquatic and terrestrial insect abundance, and tracking leaf and insect subsidies to birds and fish through isotopic measures of gut and fecal samples. In situ field manipulations such as litter and insect-emergence experiments, controlled dietary trials in grayling, long-term population surveys (through satellite imagery, mark-recapture, nest monitoring), and additional genomic datasets can reveal further aspects of how shrub evolution reshapes the flow of energy, materials, and organisms in this strongly coupled evolving meta-ecosystem.

non-heritable plasticity, an important factor to consider for species persistence (Ashander et al., 2016). Additionally, population genomic tools (e.g., genome-wide association studies and scans for selective sweeps) and landscape genomic approaches (e.g., estimating demographic history, spatial genetic structure, and relationships between environmental and genetic variation) can identify the genetic components underlying these traits while estimating the strength, timing, and spatial pattern of selection and evolution (Bernatchez et al., 2024). Similarly, comparative genomics can reveal patterns of gene family evolution, structural variation, and selective signals that may underlie adaptive strategies within lineages.

These temporal and contemporary methods quantifying organismal evolution, alongside ecosystem-level measurements or experimental manipulations—on properties such as nutrient supply and stoichiometry, productivity, matter transport rates, dissolved gases, hydrologic discharge, organismal emergence rates, movement, and migration, or trophic subsidy fluxes—allow for explicit analyses of whether genetically based trait variation alters cross-ecosystem fluxes. Establishing these links therefore requires evidence for an evolutionary response, and then tracing and quantifying its direct or indirect impact on population, community, and ecosystem dynamics throughout the meta-ecosystem—an approach outlined below for stream-riparian meta-ecosystems in the Arctic (Figure 1).

THE ARCTIC AS A TRACTABLE SYSTEM TO OBSERVE EVOLVING META-ECOSYSTEM DYNAMICS

Ecosystems characterized by relatively low species diversity and strong, well-defined coupling between ecosystems present tractable settings for developing the evolving meta-ecosystem framework. Identifying focal study organisms is

critical, but even within simplified ecosystems, this can be difficult. Thus, dominant or foundation species—including those that occupy or move across ecosystem boundaries—that mediate the abundance and flow of energy, material, and organisms offer a practical starting point (Whitham et al., 2003). Systems experiencing ongoing and rapid environmental change offer further opportunities to observe relationships between evolution and evolving meta-ecosystem dynamics.

The Arctic is well-suited for observing and understanding such dynamics. For example, the rate of Arctic warming is four times faster than global averages (Rantanen et al., 2022), leading to spatially structured selective pressures and large-scale shifts in vegetation, permafrost, hydrology, and biogeochemical cycles (Frost et al., 2025) that provide opportunities to observe evolutionary and ecological change in real time. The Arctic has a history of extensive long-term ecological research and observatory networks that provide the temporal and spatial resolution necessary to investigate multi-level dynamics. The Arctic's tundra and stream-riparian meta-ecosystems are each connected by strong coupling between biotic and abiotic processes and are composed of a few key species that have disproportionately large ecosystem impacts, where even subtle shifts in community or functional composition can trigger cascading effects at the ecosystem level (Wookey et al., 2009). The expansion and increased growth of deciduous shrubs, such as willows, birches, and alders along stream networks is one of the clearest biological signals of Arctic warming (Tape et al., 2012; Myers-Smith et al., 2020) and underscores the potential for rapid change to affect multiple dimensions of meta-ecosystem coupling.

As a dominant species at the boundary of stream-riparian zones, the widespread feltleaf willow (*Salix alaxensis* (Andersson) Coville; Salicaceae) has an outsized influence on the flow of energy, materials, and organisms across ecosystem boundaries. For example, taller, denser Arctic willows increase habitat and resources for both terrestrial insects and birds (Boelman et al., 2015) and influence levels of organic matter entering streams. Impacts to aquatic and terrestrial insect abundance may in turn affect trophic subsidies for freshwater fish such as Arctic grayling (*Thymallus arcticus* (Pallas); Salmonidae) and migratory birds such as white-crowned sparrows (*Zonotrichia leucophrys* (Forster); Passerellidae), potentially altering the flow of energy, materials, and organisms between terrestrial and aquatic ecosystems at both local and continental scales. Feltleaf willow therefore provides an excellent opportunity to understand how evolutionary changes in traits related to growth, phenology, and resource allocation could have substantial cross-system impacts (Figure 1). By revealing how evolutionary responses in dominant species can cascade across meta-ecosystem boundaries—affecting temperature regimes, nutrient fluxes, organismal abundance and diversity—Arctic stream-riparian networks provide a tractable model with principles that extend beyond polar environments, offering a translational framework for studying evolving meta-ecosystem dynamics.

SUMMARY

Darwin's tangled bank describes a local network of interconnected organisms resulting from ecological and evolutionary processes. An evolving meta-ecosystem perspective extends that metaphor to spatially dynamic interactions with consequences and feedbacks at the landscape scale, mediated by evolutionary and ecological dynamics in real time. This integrative and cross-disciplinary framework therefore offers a generalizable approach to studying eco-evolutionary dynamics in a rapidly changing world.

AUTHOR CONTRIBUTIONS

B.M.L. drafted and revised the manuscript with input and assistance from I.N.C.S., L.A.D., M.H., C.N., M.C.U., J.L.W., and P.A.H.W. Authors L.A.D., M.H., C.N., M.C.U., J.L.W., and P.A.H.W. led the conceptualization of the EVOME framework and funding acquisition.

ACKNOWLEDGMENTS

The authors are supported by the Evolving Meta-Ecosystems (EVOME) Biology Integration Institute (NSF BII 2320675). The authors would like to thank Robert Golder for the basis of the sketch used for Figure 1. We would also like to thank Chris Neill, Andrew Schnabel, and an anonymous reviewer for helpful comments that improved our manuscript.

DATA AVAILABILITY STATEMENT

No new data were created or analyzed in this study.

ORCID

Brandon M. Lind https://orcid.org/0000-0002-8560-5417
Ian N. C. Shuman https://orcid.org/0009-0008-2581-6037
Linda A. Deegan https://orcid.org/0000-0002-1017-9599
Mary Heskel https://orcid.org/0000-0003-3227-2978
Mark C. Urban https://orcid.org/0000-0003-3962-4091
Jill L. Wegrzyn https://orcid.org/0000-0001-5923-0888
Peter A. H. Westley https://orcid.org/0000-0003-4190-7314

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How to cite this article: Lind, B. M., I. N. C. Shuman, L. A. Deegan, M. Heskel, M. C. Urban, J. L. Wegrzyn, and P. A. H. Westley. 2026. Beyond the tangled bank: An evolving meta-ecosystems framework to integrate ecological and evolutionary perspectives. *American Journal of Botany* 113: e70274. https://doi.org/10.1002/ajb2.70274